17 From Probability to Statistics

Example 17.1. In each of the following scenarios, one of (i) and (ii) is a *probability* and one is a *statistics* question. Classify each question as "probability" or "statistics". Then discuss some general features of the probability questions, and some general features of the statistics questions

1. U.S. Satisfaction
   - Assume that 30% of Americans are satisfied with the way things are going in the U.S. If a random sample of 1000 Americans is selected, what are the chances that more than 330 Americans in the sample are satisfied with the way things are going in the U.S?
   - In a random sample of 1000 Americans, 330 Americans are satisfied with the way things are going in the U.S. What percent of all Americans are satisfied with the way things are going in the U.S?
2. Great white shark length
   - Assume that lengths of female great white sharks follow a Normal distribution with mean 16 ft and standard deviation 3 ft. If a random sample of 40 female great white sharks is selected, what are the chances that the average length in the sample is between 16.5 and 17.5 ft?
   - In a sample of 40 female great white sharks, the sample mean length is 17.0 ft. What is the average length of female great white sharks?
3. In a clinical trial, 200 subjects are randomly assigned to receive either an experimental vaccine or no treatment (e.g., a placebo), and then each subject is tested to see if they developed immunity.
   - If the vaccine is in reality not effective, what are the chances that the proportion of subjects who develop immunity is twice as large for those who received the vaccine than for those who did not?
   - Among the 100 subjects who received the vaccine, 20 developed immunity; among the 100 subjects who did not receive the vaccine, 10 developed immunity. Is there evidence that the vaccine is effective?

4. Snap streaks

    - Assume that lengths of streaks on Snapchat follow an Exponential distribution, with mean 75 for teen users and mean 40 for adult users. A random sample of 200 Snapchat users is selected, 100 teens and 100 adults. What are the chances that the sample mean streak length for teens is with within 20 of the sample mean streak length for adults?
    - In a random sample of 100 teen Snapchat users, the sample mean streak length is 80. In a random sample of 100 adult Snapchat users, the sample mean streak length is 35. In general, how much longer do streak lengths for teen Snapchat users tend to be than for adult Snapchat users?

- **Probability:** Assume a model for a random (uncertain) process and evaluate probabilities of potential outcomes or events — what might the data look like?
    - In probability, a model is assumed and parameters are assumed to be known, but the data is unknown.
    - In probability, we typically call the model a "distribution" and the data "random variables".
- **Statistical inference:** Observe sample data and make conclusions about the process that generated the data
    - In statistics, the data is observed but parameters are unknown.
    - In statistics, we typically call the model the "population" and the data the "sample".
- *Statistical inference* involves using sample data to make conclusions about the population.
    - **Point estimation:** Estimate an unknown population parameter with a single number based on sample data
    - **Interval estimation:** Estimate an unknown population parameter with a plausible range of values based on sample data

- **Hypothesis testing:** Assess the evidence that the sample data provides regarding a particular claim about unknown population parameters.

- The *population distribution* is the distribution of individual values of the variable of interest. A *parameter*, generically denoted θ, is a number that summarizes the population distribution.

- The **population distribution**, denoted p_θ or $p_\theta(x)$, is a probability model for the distribution of values of a variable[1] X over all *individuals* in the population.

  - If the variable X is discrete, then p_θ represents a probability mass function (pmf): $p_\theta(x) = \mathrm{P}(X = x)$

  - If the variable X is continuous, then p_θ represents a probability density function (pdf): $p_\theta(x) \approx \frac{1}{\epsilon}\mathrm{P}(x - \epsilon/2 < X < x + \epsilon/2)$

- A **parameter**, generically denoted θ, is a characteristic of the population distribution, often computed as an expected value of some function of X.

  - In some cases θ represents a vector of multiple parameters.

  - For example, if p_θ represents a Normal distribution, then θ might represent the pair $\theta \equiv (\mu, \sigma)$ consisting of the population mean and the population standard deviation.

- A parameter is a *fixed number* but its value is *unknown*[2].

- Many statistical studies involve *random sampling* from a population.

- A **(simple) random sample** of size n is a collection of random variables $X_1, \dots, X_n$ that are *independent* and *identically distributed* (i.i.d.)

  - Identically distributed — each individual X_i has the same marginal distribution — the population distribution — since they are sampled from the same population:

$$X_i \sim p_\theta \quad \text{for all } i = 1, \dots, n$$

[1]We usually have i.i.d. observations and so the joint density is just the product of the marginal densities as we have defined it. However, for more general situations the likelihood can be defined directly as the joint density.

[2]There are two main philosophical approaches to statistical inference that differ mainly in whether they emphasize "fixed number" or "unknown". The *frequentist* approach treats a parameter as a *fixed number*. The *Bayesian* approach treats an *unknown* parameter— in other words, an *uncertain* parameter—as a *random variable* with a probability distribution that describes the degree of uncertainty. Both approaches are valid, each with advantages and disadvantages. The frequentist approach has been historically more prevalent, but the Bayesian approach has gained in popularity in the last 30 or so years.

- Independent — the values are sampled independently, so the joint distribution[3] is the product of marginal distributions:

$$p_\theta(x_1, \dots, x_n) = p_\theta(x_1)p_\theta(x_2)\cdots p_\theta(x_n) \quad \text{for all } x_1, \dots, x_n$$

- A **statistic** is a characteristic of the *sample* which can be computed from the data. More precisely, a statistic is a function of $X_1, \dots, X_n$, but not of θ.
  - That is, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
  - The probability distribution of a statistic is called a **sampling distribution**.
  - The *distribution* of a statistic will depend on θ, but the *value* of a statistic will not.
- A statistic $\hat{\theta} \equiv T(X_1, \dots, X_n)$ used to estimate a parameter θ is called a **(point) estimator**.
  - Naturally, we hope that there is some correspondence between our estimator $\hat{\theta}$ and the parameter it is estimating θ — for example, estimating the population mean with the sample mean — but the definition does not require this.
  - When the numerical value of a statistic is calculated from observed sample data, its value is called a *(point) estimate* of the parameter.
  - The estimator $\hat{\theta}$ is a RV whose value changes from sample to sample; an estimate is the observed value computed based on the results of a single sample.

Example 17.2. Jerry has just opened a car dealership in a new location. He assumes that the number of cars he'll sell in a day has a Poisson(μ) distribution. But he doesn't know the value of μ, so he plans to use data from his first few days of business to estimate μ. Let $X_1, \dots, X_n$ represent the number of cars sold for a sample of n days.

1. What plays the role of the population distribution? What is the parameter? How do you interpret the parameter in this context?

[3] Abuse of notation in the expression: p_θ on the left represents the joint distribution of $(X_1, \dots, X_n)$, a function with n inputs; p_θ on the right represents that marginal distribution of each X_i (since i.d.), a function with a single input.

2. Suggest one estimator of μ. (Hint: what does μ represent for a Poisson distribution?)

3. Suppose he collects data on $n = 3$ days and observes $x_1 = 3$ cars sold on day 1, $x_2 = 0$ cars sold on day 2, and $x_3 = 2$ cars sold on day 3. Compute the value of your estimator from the previous part for this sample. That is, what is your estimate of μ based on this sample?

4. Suggest another estimator of μ. (Hint: what else does μ represent for a Poisson distribution?)

5. For a Poisson(μ) distribution μ is both the population mean and the population variance. So one reasonable estimator of μ is the sample mean $\bar{X}$. Another reasonable estimator is the sample variance. But there are two commonly used formulas for the sample variance, each of which provides an estimator of μ

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$
$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

   Compute the value of each of these estimators for the (3, 0, 2) sample. That is, what is your estimate of μ based on this sample if you use each of these estimators?

6. At his previous dealeship at a different location, Jerry sold 2.3 cars per day on average. Consider the estimator of μ: $\frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3)$. Explain in words what this estimator represents. In particular, what happens as n increases? Then compute the value of this estimator for the (3, 0, 2) sample.

7. According to the Poisson(μ) distribution, the probability of selling 0 cars in a day is $p_0 = e^{-\mu}$. Rearranging, $\mu = -\log p_0$. Therefore, if $\hat{p}_0$ is the sample proportion of days with 0 cars sold, then another estimator of μ is $-\log \hat{p}_0$. Compute the value of this estimator for the (3, 0, 2) sample.

8. We have seen a few different, seemingly reasonable estimators of μ, and each produced a different estimate of μ based on the same (3, 0, 2) sample data. Which of these numbers is the best *estimate* of μ? That is, which of these estimates is closest to μ?

9. How might you investigate which of the *estimators* corresponds to the best estimation *procedure*?

- Because the true value of the parameter θ is unknown, we never know if the sample data provides a reasonable estimate of θ
- Rather, we evaluate the estimation *procedure* for potential values of θ: How does the estimation procedure perform over many possible samples given potential values of θ?
- Some questions of interest when estimating a parameter θ
  - How do we find an estimator of θ?
  - What are properties of "good" estimators?
  - How can we tell if one estimator is "better" than another?
  - Can we find a "best" estimator of θ? (Short answer: no)
  - How do we determine the margin of error for an estimator?

18 Maximum Likelihood Estimation: Principles

Example 18.1. Suppose we are willing to assume a Poisson(μ) model for the number of cars sold in a day at a particular car dealership. But we don't know the value of μ, so we'll collect data and use it to estimate μ. Suppose first that we observe just a single day; let X be the number of cars sold on this day. (We'll look at a sample of n days soon.)

1. Does X take values on a discrete or continuous scale? What about μ?

2. *First some probability questions.* Assume $\mu = 2.3$. Find the pmf of X and sketch a plot of it; what is this a function of? Find and interpret the probability that in a single day there are 3 cars sold.

3. *Now back to some statistics questions.* Now remember that in reality μ is unknown. Suppose that $x = 3$ home runs are observed in a single day. Now write the pmf plugging in everything we know; what is this a function of? Is this function a pmf?

4. Compute and intepret the probability that in a single day there are 3 cars sold if $\mu = 2.3$.

5. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 1$.

6. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 5$.

7. Compute and interpret the probability that in a single day there are 3 cars sold if $\mu = 3.5$.

8. Suppose that $x = 3$ cars are sold in a single day. Based just on this day, which value — 1, 2.3, 3.5, or 5 — would you choose as your estimate for μ? Why?

9. We obviously have more choices for our estimate of μ than just 1, 2.3, 3.5, or 5. Describe in principle the process you would follow to find the estimate of μ based on a single day with 3 cars sold.

10. Suppose that we observe $x = 3$ cars sold in a single day. Plot the likelihood function; be sure to label your axes. Based on your plot, what would be your estimate of μ?

11. Suppose that we observe $x = 3$ cars sold in a single day. Plot the log of the likelihood function. Compare the plots of the likelihood function and the log-likelihood; what do

you notice about where the maximum value occurs?

12. Now consider a general value of x cars solds in a single day. Carefully write the likelihood function, and the log-likelihood function. What are these functions of? How would you use them to determine your estimate of μ given x cars sold in a single day?

- Consider a single observation X from a population p_θ with parameter θ: Let $X \sim p_\theta$. The **likelihood function** given $X = x$ is defined as

$$L(\theta) \equiv L_x(\theta) = p_\theta(x)$$

- That is, the likelihood function is the probability (or density for continuous data X) of observing the given sample data x viewed as a *function of the unknown parameter(s)* θ. The observed value of x (data) is treated as a fixed constant.
- The *maximum likelihood estimate* of θ is the value of θ for which the observed data has the highest likelihood of occurring.
- The **maximum likelihood estimator (MLE)** of θ is the RV $\hat{\theta}$ which maximizes the likelihood, i.e. the value $\hat{\theta}$ which satisfies

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta.$$

- Remember that an estimator $\hat{\theta}$ is a function of X, the observed data.
- It is often convenient to consider the **log-likelihood function** of θ

$$\ell(\theta) \equiv \ell_x(\theta) = \log L_x(\theta)$$

- Because log is an increasing function it follows that $\hat{\theta}$ is the MLE of θ if and only if

$$\ell(\hat{\theta}) \geq \ell(\theta) \text{ for all } \theta.$$

Example 18.2. Suppose we are willing to assume a Poisson(μ) model for the number of cars sold in a day at a particular car dealership. But we don't know the value of μ, so we'll collect data and use it to estimate μ. But now we estimate μ based on a sample of n days. Let $X_1, \dots, X_n$ be the number of cars sold in a random sample of n days. For example, suppose $n = 3$ and we observe $x_1 = 3$, $x_2 = 0$, $x_3 = 2$.

1. *First some probability questions.* Assume $\mu = 2.3$ and $n = 3$. Find the joint pmf of (X_1, X_2, X_3); what is this a function of? Find and interpret the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$.

2. *Now back to some statistics questions.* Now remember that in reality μ is unknown. Suppose that the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ is observed. Now write the joint pmf plugging in everything we know; what is this a function of? Is this function a pmf?

3. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 2.3$.

4. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 1$.

5. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 5$.

6. Compute the probability of observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ if $\mu = 3.5$.

7. Suppose the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$ is observed. Which value — 1, 2.3, 3.5, or 5 — would you choose as your estimate for μ? Why?

8. We obviously have more choices for our estimate of μ than just 1, 2.3, 3.5, or 5. Describe in principle the process you would follow to find the estimate of μ based on observing the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$.

9. Suppose that we observe the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$. Plot the likelihood function; be sure to label your axes. Based on your plot, what would be your estimate of μ?

10. Suppose that we observe the sample $(x_1 = 3, x_2 = 0, x_3 = 2)$. Plot the log of the likelihood function. Compare the plots of the likelihood function and the log-likelihood; what do you notice about where the maximum value occurs?

11. Now consider a general sample of size n. Carefully write the likelihood function, and the log-likelihood function. What are these functions of? How would you use them to determine your estimate of μ given x cars sold in a single day?

- Consider a random sample of n values from a population p_θ with parameter θ: Let $X_1, \dots, X_n$ be i.i.d. $\sim p_\theta$.
- The **likelihood function** of θ, given $X_1 = x_1, \dots, X_n = x_n$, is

$$L(\theta) \equiv L_{x_1,\dots,x_n}(\theta) = \prod_{i=1}^{n} p_\theta(x_i) = p_\theta(x_1) \times \cdots \times p_\theta(x_n)$$

- That is, the likelihood function is the joint density[1] of the sample $X_1, \dots, X_n$, evaluated at the observed data values $x_1, \dots, x_n$, viewed as *a function of the (unknown) parameter(s) θ.*
- Note that $x_1, \dots, x_n$ represent distinct values (like $x_1 = 3, x_2 = 0, x_3 = 2$ in the example).
- The MLE of θ is the value of θ for which the observed sample data has the highest likelihood of occurring.
- The **maximum likelihood estimator (MLE)** of θ is the RV $\hat{\theta}$ which maximizes the likelihood, i.e. the value $\hat{\theta}$ which satisfies

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta.$$

- Remember that an estimator $\hat{\theta}$ is a function of $X_1, \dots, X_n$.
- The **log-likelihood function** of θ is

$$\begin{aligned}\ell(\theta) \equiv \ell_{x_1,\dots,x_n}(\theta) &= \log L(\theta) \\ &= \sum_{i=1}^{n} \log p_\theta(x_i) = \log p_\theta(x_1) + \cdots + \log p_\theta(x_n)\end{aligned}$$

- Because log is an increasing function it follows that $\hat{\theta}$ is the MLE of θ if and only if

$$\ell(\hat{\theta}) \geq \ell(\theta) \text{ for all } \theta.$$

Example 18.3. Suppose instead of observing the number of cars sold on individual days, you only knew that in a sample of 3 days there were 5 cars sold in total.

1. *Probability question.* If $X_1, \dots, X_n$ are i.i.d. Poisson(μ), what is the distribution of $X_1 + \cdots + X_n$?

2. Write the likelihood function for a sample of 3 days in which a total of 5 cars were sold. How does this compare to the likelihood function based on the sample $(3, 0, 2)$?

[1]We usually have i.i.d. observations and so the joint density is just the product of the marginal densities as we have defined it. However, for more general situations the likelihood can be defined directly as the joint density.

3. Find the MLE of μ based on a sample of 3 days in which a total of 5 cars are sold.

- In many situations, the full sample data is not necessary to find the MLE; rather, a few descriptive statistics are often sufficient to evaluate the likelihood and determine the MLE
- Poisson(μ): the MLE of μ is the sample mean $\bar{X}$
- Binomial(n, p), with n known: the MLE of p is the sample proportion $\hat{p} = X/n$, where X is the number of "successes" in the sample
- Exponential(λ): the MLE of λ is the sample rate $1/\bar{X}$; the usual situation is where the X's measure times between "events" and so $\bar{X}$ is the sample mean time between "events".
- Normal(μ, σ), with both μ and σ unknown: the MLEs of μ, σ^2, and σ, are, respectively

$$\hat{\mu} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

$$\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2}$$

Example 18.4. Assume a Poisson(μ) model for the number of cars sold in a day at a particular car dealership. In 5 days, 3 cars are sold.

1. Consider $\sqrt{\mu}$. Is $\sqrt{\mu}$ a parameter or a statistic? How would you interpret $\sqrt{\mu}$ in this context?

2. What do you think the MLE of $\sqrt{\mu}$ is?

3. Consider $e^{-\mu}$. Is $e^{-\mu}$ a parameter or a statistic? How would you interpret $e^{-\mu}$ in this context?

4. What do you think the MLE of $e^{-\mu}$ is?

5. Consider $\mu e^{-\mu}$. Is $\mu e^{-\mu}$ a parameter or a statistic? How would you interpret $\mu e^{-\mu}$ in this context?

6. What do you think the MLE of $\mu e^{-\mu}$ is?

7. Consider $\mu^2 e^{-\mu}/2$. Is $\mu^2 e^{-\mu}/2$ a parameter or a statistic? How would you interpret $\mu^2 e^{-\mu}/2$ in this context?

8. What do you think the MLE of $\mu^2 e^{-\mu}/2$ is?

9. We started by assuming a Poisson model, but how do we know that is a reasonable assumption? Suppose we observe the number of cars sold for a random sample of days. Suggest you might use the sample data to investigate whether a Poisson model is appropriate.

- A parameter is any characteristic of a population distribution
- For many populations, there are a few "main" parameters (like μ for Poisson(μ), or (μ, σ) for Normal(μ, σ), but any characteristic of a population is parameter and can be estimated
- **Invariance property of MLEs.** If $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$.
- That is, the "MLE of a function is the function of the MLE"
- The invariance principle says that if you find the MLE of the "main parameters" then you automatically get the MLE of all the parameters that are based on it

Example 18.5. Suppose that within a certain population, credit scores follow a Normal(μ, σ) distribution. In a random sample of 5 individuals from this population, the credit scores are 600, 630, 680, 700, 770.

1. Compute the MLE of μ.

2. Compute the MLE of σ.

3. Explain in words what it means for these parameters to be the MLE.

4. Compute the MLE of the 16th percentile of credit scores for this population

5. Compute the MLE of the 90th percentile of credit scores for this population

6. Compute of the MLE of the population proportion of individuals with a credit score above 750.

- Maximum likelihood estimation procedures are widely applicable and maximum likelihood estimators have many nice theoretical properties.
- However, there are some drawbacks:
  - May require numerical maximization, which can be computationally intensive (especially if many parameters)
  - May not be reliable estimators in small samples. (Many of the nice theoretical properties of MLEs are true for large samples.)

19 Maximum Likelihood Estimation: Calculus

Example 19.1. We wish to estimate the parameter μ for a Poisson(μ) distribution based on a single observed value X.

1. Suppose that $x = 3$. Carefully write the likelihood function.

2. Suppose that $x = 3$. Carefully write the log-likelihood function.

3. Use calculus to find the MLE of μ given $x = 3$. Compare to what we found previously using graphical methods.

4. Now consider a general x. Carefully write the likelihood function.

5. Now consider a general x. Carefully write the log-likelihood function.

6. Use calculus to find the MLE of μ given x. Compare to what we found previously using graphical methods.

Example 19.2. We wish to estimate the parameter μ for a Poisson(μ) distribution based on a random sample of n values $X_1 \dots, X_n$.

1. Suppose that $n = 3$ and the sample is (3, 0, 2). Carefully write the likelihood function.

2. Suppose that $n = 3$ and the sample is (3, 0, 2). Carefully write the log-likelihood function.

3. Use calculus to find the MLE of μ given $n = 3$ and the sample (3, 0, 2). Compare to what we found previously using graphical methods.

4. Now consider a general sample of size n. Carefully write the likelihood function.

5. Now consider a general sample of size n. Carefully write the log-likelihood function.

6. Use calculus to find the MLE of μ given a sample of size n. Compare to what we found previously using graphical methods.